

Performance Report

Evaluation

Installation	Equipment	Type	No.
APL HOUSTON	MAIN ENGINE	10S90ME-C	DML0105356

Readings Included

	Date	Voyage / title	Remarks
1	2025-10-25	04FLZW1MA	ENROUTE FROM NANSHA TO SINGAPORE

Main indications (all Readings)

Reading	1
Description	Most recent
Fuel oil saving potential	●
Light running	●
Fuel index ECU	●
Firing press. AVG	●
Compression press. AVG	●
Scav. air press. RECEIVER	●
Exh. gas temp. TC in AVG	●
Exh. gas temp. TC out AVG	●
TC speed AVG	●
Spec. FO consumption	●

Indications (for Reading 1)

Description	Indication	Comments / recommendations
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Optimization potential

Quantification						
Influenced by	Fuel oil equiv.	Thermal load	Proposed actions to achieve the saving / reduction			
Firing pressure by injection	0.0 g/kWh	0 °C				
Firing pressure by compression ratio	0.2 g/kWh	0 °C	Increase compress. press. (compr. ratio offset) to raise firing press. by 1 bar. Or advance inj. begin to compensate.			
Firing pressure by scavenge air pressure	0.0 g/kWh	0 °C				
			Target firing press.: 145 bar (ISO corr., REF.) or 145 bar (meas., AVG.). Check OEM limits pmax - pcomp: 35 bar.			
ENGINE ROOM ventilation	0.0 g/kWh	0 °C				
Suction press. air TC in	0.0 g/kWh	0 °C				
Press. drop accross SAC	0.0 g/kWh	0 °C				
Scav. air : water temp. SAC in	0.0 g/kWh	0 °C				
Scav. air : temp. RECEIVER	0.1 g/kWh	1 °C	Increase water flow, clean SAC water- and/or air side to reduce scav. air temp. by 1 °C to 39 °C.*			
Press. drop accross ENGINE	0.6 g/kWh	9 °C	Air- or gas flow restrictions: High press. drop across SAC / Exh. gas grid TC in clogged / Clean LINER inlet ports.			
Exh. gas press. TC out	0.0 g/kWh	0 °C				
Light running	0.0 g/kWh	-				
Fuel oil visc. ENGINE in	0.0 g/kWh	0 °C				
	Fuel oil equiv.	Thermal load	Cyl. lub. oil			
TOTAL (specific)	0.9 g/kWh	-				
TOTAL (of current)	0.6 %	-		CO2 (emission)	SOx (emission)	
TOTAL optimization potential	0.58 t/d	10 °C		1.8 t/d	6 kg/d	
	Cost					
Fuel oil price						
TOTAL saving potential						

Consumption

Measured / calculated values			
Indication	Fuel oil		Cylinder lub oil
Specific consumption reference	166.2 g/kWh		0.70 g/kWh
Specific consumption ISO / MEAS	172.9 g/kWh		0.71 g/kWh
Consumption per hour day *	4571 kg/h	109.7 t/d	18.1 kg/h
Consumption nautical *	246 kg/nmi		
* Measured values, uncorrected			

Injection timing and compression ratio adjustment

[illegible]

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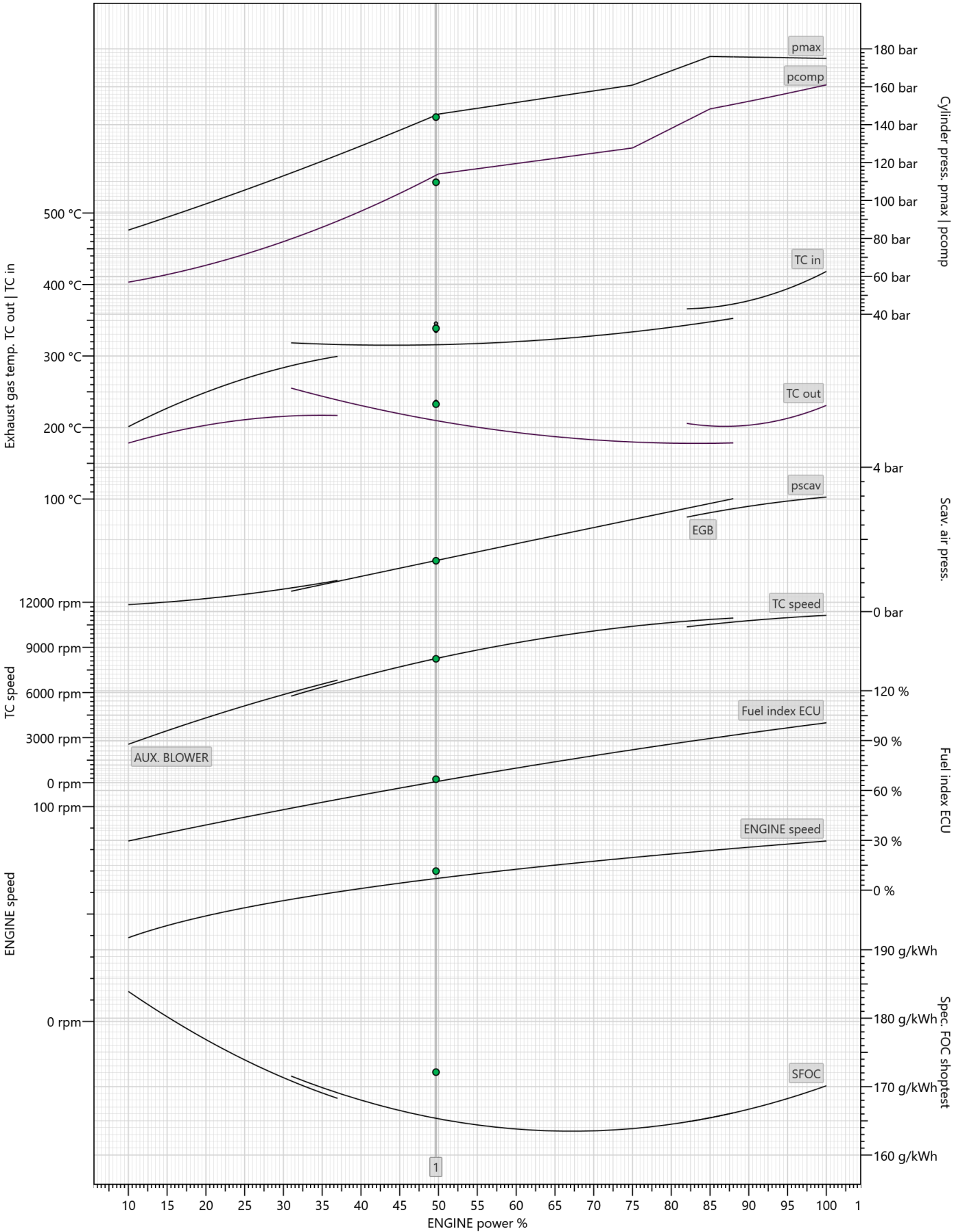
Input plausibility

Value	Comment
Compressor evaluation	
Exhaust gas pressure Manifold	
PTI	
PCO	
TTI vs TTO	
TCO	
Minimum required data	
Minimum required data II	
Missing data	
Missing data II	

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Performance Curves

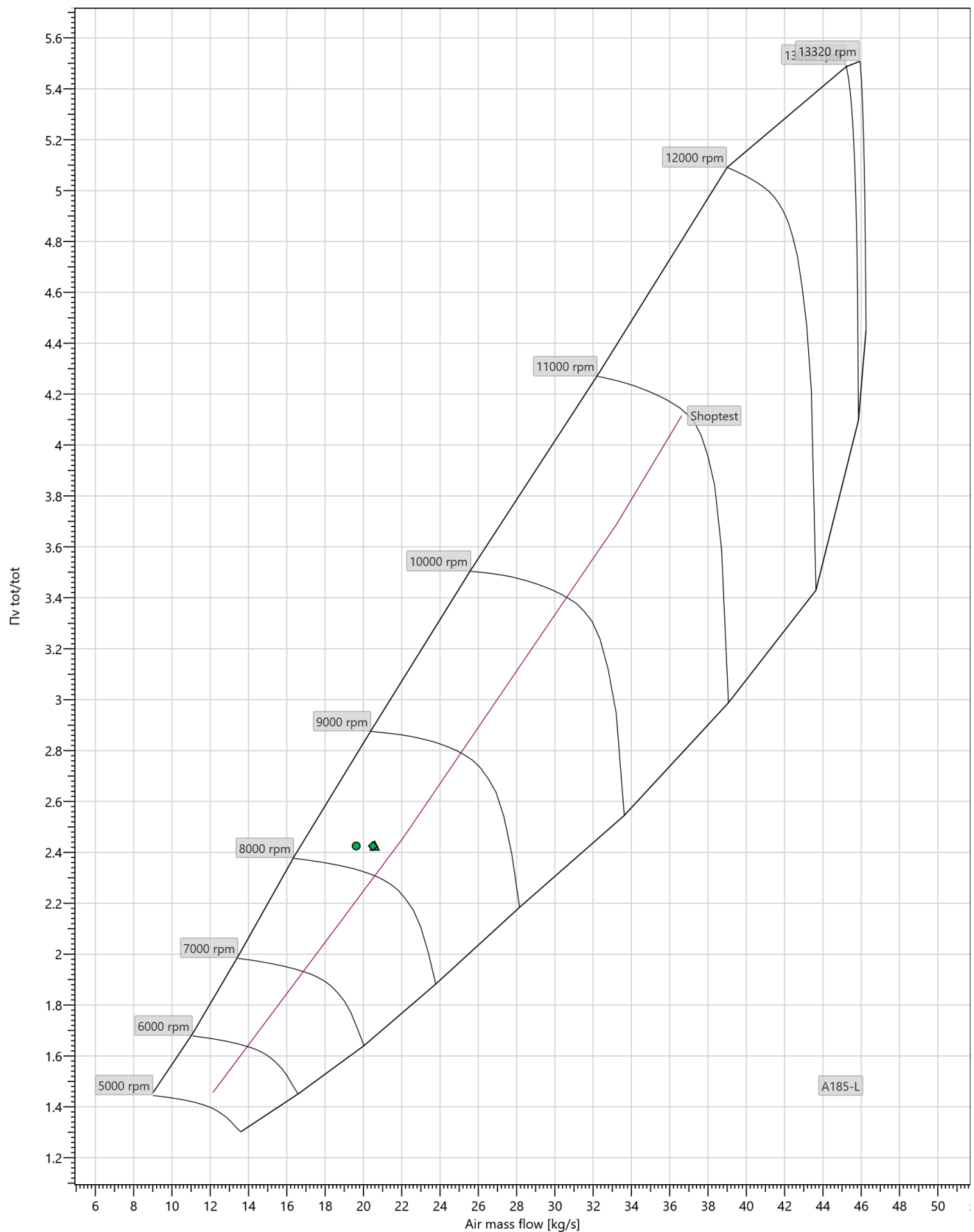
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COMPRESSOR map

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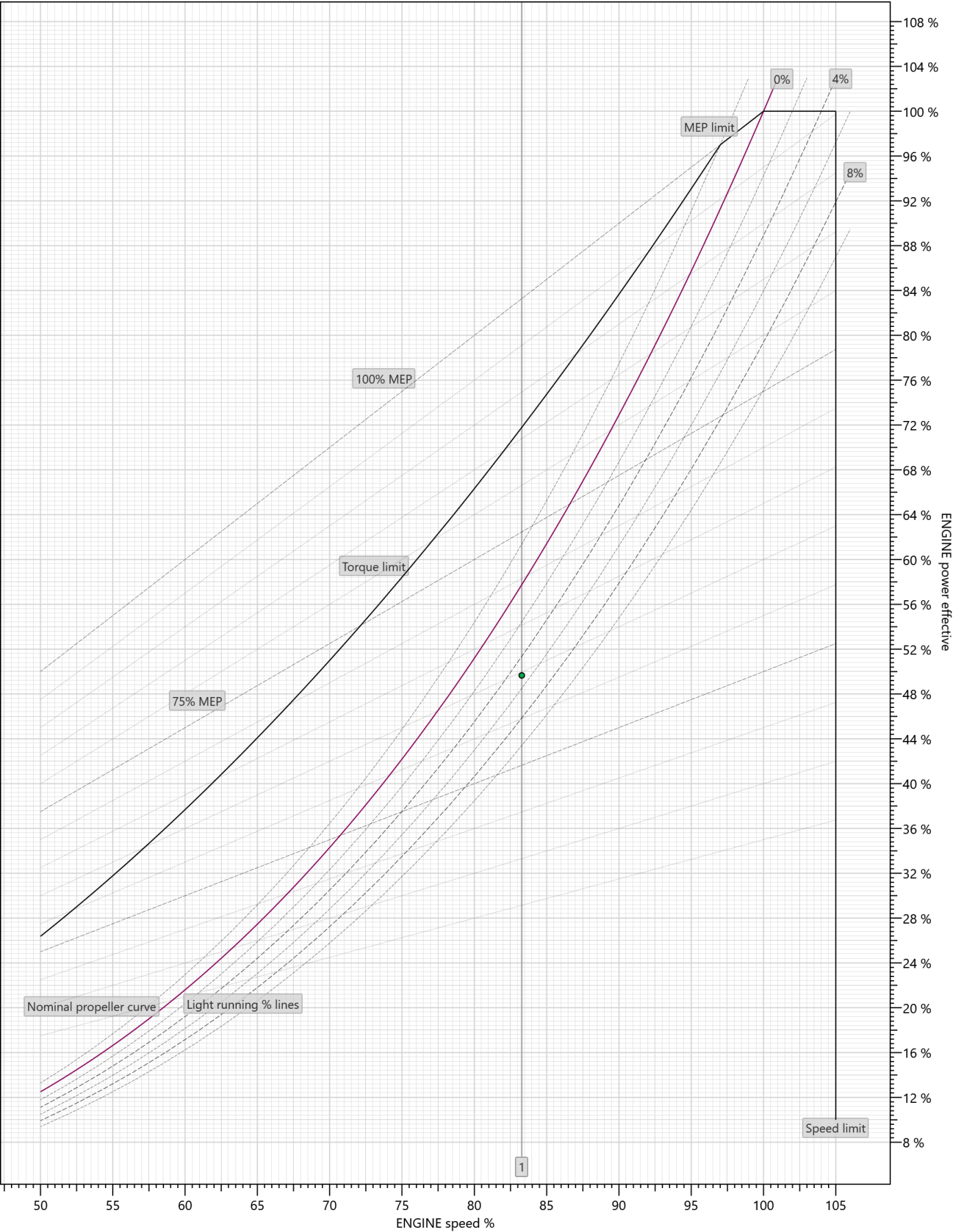


1	2025-10-25 04FLZW1MA		
● TC 1	▲ TC 2	◆ TC 3	

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Load Diagram

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Reading

Installation			Equipment			Type			No.																					
APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356																					
General	Date and time of recording		10/25/2025 15 : 08 : 50		Sea water temp.		29.0 °C		SHIP SOG STW		19.30 18.60 kn		Voy. / title		04FLZW1MA															
	ENGINE state		stable		Ambient air temp.		29.0 °C		Draft fwd aft		12.60 12.80 m		Record by		2/E															
	ENGINE running hours		69556 h		Engine room air temp.		30.0 °C		Draft mid Trim		12.70 0.20 m		Ackn. by		C/E															
	Ambient humidity		rel 32.0 % abs 8.0 g/kg		Barom. pr. Eng. room		1.010 1.013 barA		Remarks		ENROUTE FROM NANSHA TO SINGAPORE																			
Power / Speed			CALCULATED		MEAS		ESTIMATED 24.3(E264X126)																							
			XPERT		TORSIOM.		MOP		by FI*rpm		by pscav		by TC rpm		by MEP		by rpm													
	ENGINE power estimated		kW 25357 % 49.7		25730		25535		25886		25311		25339		24834															
	ENGINE speed		rpm 84.0 83.3 % 70.0		ENGINE		TORSIOMETER		MARGINS		MECH. EFFICIENCY																			
	ENGINE power effective		kW 51070 49.7 % 25357				Power kW 25730		Light running 5.2 %		Theoretic. 93.8 %																			
(for PTI operation, use negative value, "-")																														
Electronic ctrl	Running mode		Economy		ISO CORRECTED		MEASURED																							
	Governor mode		RPM		REF.		CALC.		ALL																					
	Fuel index ECU		% 65.5 66.8		63.0																									
	Applied fuel quality offset		% 0.0																											
Cylinder pressure	SOURCE: MAN CoCoS		ISO CORRECTED		MEASURED												SETPOINT													
	QUALITY Imported		REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10		MOP	
	Firing press. pmax		barG 145.0		144.0		144.0		144.0		144.0		144.0		144.1		144.1		144.0		144.0		144.0		144.0		144.0			
	pmax deviation		bar		ALL		0.0		0.0		0.0		0.0		0.0		0.1		0.1		0.0		0.0		0.0		0.0			
	pmax offset		bar		MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Compression press.		barG 113.3		109.6		109.7		109.7		109.6		109.7		109.6		109.6		109.7		109.7		109.7		109.6		109.6			
	pcomp deviation		bar		ALL		0.1		-0.1		0.1		-0.1		-0.1		0.1		0.1		0.1		-0.1		-0.1		-0.1			
	pcomp / pscav offset		-		MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Mean indicated press.		barG 11.16		10.95		11.40		10.90		10.90		10.90		10.90		10.90		10.90		10.90		10.90		10.90		10.90			
	MIP deviation		bar				0.45		-0.05		-0.05		-0.05		-0.05		-0.05		-0.05		-0.05		-0.05		-0.05		-0.05			
	Fuel offset high		%		MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Fuel offset low		%		MOP SETTING:		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0		0.0			
	Power indicated		kW		SUM:		26479		2757		2636		2636		2636		2636		2636		2636		2636		2636		2636			
	Mean effective press. (MEP)		bar 10.49		10.27		58.4 %		of MCR																					
	pmax-pcomp pcomp / pscav		- 31.7 47.4		34.4 45.9														pcomp / pscav		48.1									
	Turbocharger			ISO CORRECTED		MEASURED																								
				REF.		CALC.		AVG. / ENG.		TC 1		TC 2		TC 3																
PV (tot-tot)								2.42		2.43		2.42																		
TC speed		rpm 8257		8241		8280		8244		8305		8290																		
TC speed 298		rpm 8234		8200		8166		8131		8191		8177																		
Turbocharger efficiency																														
Surge margin																														
Air mass flow		kg/s 65.1		60.5		59.5		19.2		20.2		20.1																		
Break spec.air cons.		kg/kWh		8.4																										
Exhaust gas mass flow		kg/s 65.6		61.2		60.1		19.5		20.4		20.3																		
Scavenge air			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		TC 1		TC 2		TC 3																	
	AUX. BLOWER operation		off																											
	Suction press.		mmWG 25		27		27		27		27																			
	Press. drop accr. SAC		mmWG 149		143		145		140		145																			
	Scav. air press. RECEIVER		barG 1.41		1.41		1.40																							
	Scav.air press.setp. MOP		barG																											
	Air temp. COMPR. in		°C		32.0		32.0		32.0		32.0																			
	Press. COMPR. outlet *		barG 1.41		1.40		1.40		1.40		1.40																			
	Scav. air temp. COMPR. out		°C 136		137		137		138		137																			
	Scav. air temp. SAC out		°C		38.0		38.0		39.0																					
	Scav. air temp. RECEIVER		°C 38.7		40.0																									
	Cool. water temp. SAC in		°C 35		34.0		34.0		34.0		34.0																			
	Cool. water temp. SAC out		°C		43.0		43.0		43.0		43.0																			
	*measured at flange																													
Exhaust gas			ISO CORRECTED		MEASURED																									
			REF.		CALC.		AVG.		CYL 1		CYL 2		CYL 3		CYL 4		CYL 5		CYL 6		CYL 7		CYL 8		CYL 9		CYL 10			
	Exh. gas temp. CYL out		°C 283		280		295		285		290		295		300		285		298		300		298		295		300			
	temp. deviation		°C		ALL		-10		-5		0		5		-10		3		5		3		0		5					
	Exh. vlv. open offset		°CA		MOP SETTING:																									
	Exh. gas press. MANIFOLD		barG 1.28		1.21		1.20		TC 1		TC 2		TC 3																	
	Exh. gas press. TC in *		barG 1.25		1.20		1.20		1.20		1.20																			
	Exh. gas press. TC out		mmWG 74		75		74		75		75																			
	Exh. gas temp. TC in		°C 316		339		353		350		360		350																	
	Exh. gas temp. TC out		°C 210		233		247		245		250		245																	
	Exh.wastegate (EGB) pos. area %		0 0 0 0																											
	*measured at flange																													

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APL HOUSTON			MAIN ENGINE			10S90ME-C			DML0105356							
Fuel oil			ISO CORRECTED		MEAS	Fuel oil properties:		ANALYS.	ESTIMATE	Fuel oil system:		REF	MEAS	EXPECT.		
	Absolute consumption	m³/h ▾	REF.	CALC.	5.000	Viscosity	cSt	250	@ 50 °C	Viscosity ENGINE in	cSt	12.5	12.0	13.9		
	Specific FOC	g/kWh	166.2	172.9	180.3	Sulfur content	%	0.50	0.50	Temp. ENGINE in	°C	135	130			
	Light running correction	g/kWh	0.0			Water content	%	0.20	0.20	Temp. FLOWMETER	°C		90			
	Thrust bearing loss	g/kWh	-0.8			LCV	kJ/kg	41410	41410	Density FLOWM.	kg/m³		914.2			
	Specific FOC shoptest	g/kWh	165.4	172.1		Density	kg/m³	962.0	@ 15 °C	Price	USD/t					
						Source of analysis	Laboratory ▾									
Cyl. lub.			REFERENCE		MEAS	Cyl. lub. system (Alpha):		REF	SET**	Cyl. lub. oil:		REF	ANALYS.			
	Absolute consumption	l/h ▾			19.6	Target feed rate@MCR	g/kWh	0.60	0.70	BN		40 - 70	70			
	Spec. lub oil consumpt. *	g/kWh			0.71	Target f.r.@current load	g/kWh	0.70	SET based	Density	kg/m³		940.0	@ 15 °C		
	* Effective feed rate					** Acc.to OEM operation manuals and/or drainoil analysis					kg/m³		924	@ 40 °C		
Cylinder condition			REFERENCE		MEASURED											
			REF.	AVG. ENGINE / CYL 1												
	Cool. water temp. CYL in	°C	75	80.0	80	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10		
	Cool. water temp. CYL out	°C	90	86.9	87	88	87	88	85	88	87	86	88	85		
	Cooling water press. CYL in	bar		5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0	5.0		
	Coolant temp. PISTON in	°C		45	45	45	45	45	45	45	45	45	45	45		
	Coolant temp. PISTON out	°C		56	56	56	56	55	55	56	56	56	55	56		
Scav. air temp. PIST. UNDERS.	°C		49.2	49	47	49	51	48	50	51	50	49	48			
Crankcase			MEASURED													
			AVG. / ENG.	BRG 1	BRG 2	BRG 3	BRG 4	BRG 5	BRG 6	BRG 7	BRG 8	BRG 9	BRG 10	BRG 11	BRG 12	BRG 13
	MAIN BEARING temp.	°C	54.2	54.0	53.0	56.0	56.0	55.0	55.0	53.0	52.0	53.0	53.0	56.0	57.0	51.0
	Temp. deviation	°C		0	-1	2	2	1	1	-1	-2	-1	-1	2	3	-3
			CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10				
	CRANK PIN BEAR. temp	°C	51.5	51.0	52.0	51.0	51.0	52.0	52.0	51.0	52.0	51.0	52.0			
	Temp. deviation	°C		-1	1	-1	-1	1	1	-1	1	-1	1			
X-HEAD BEAR. temp	°C	50.6	48.0	50.0	51.0	51.0	51.0	51.0	51.0	51.0	51.0	51.0				
Temp. deviation	°C		-3	-1	0	0	0	0	0	0	0	0	0			
Liner wall			MEASURED													
			CYL 1	CYL 2	CYL 3	CYL 4	CYL 5	CYL 6	CYL 7	CYL 8	CYL 9	CYL 10				
	LINER WALL temp. aft	°C														
	LINER WALL temp. fp	°C		126	125	134	135	142	131	130	123	134	135			
	LINER WALL temp. fwd	°C														
LINER WALL temp. exh	°C	AVG.	125	133	139	135	138	136	135	135	132	136				
LINER WALL temp. AVG	°C	133														
TC/SAC media			MEASURED													
			AVG.	TC 1	TC 2	TC 3										
	Water press. SAC in	bar	2.9	2.9	2.9	2.9										
	Lub oil temp. TC in	°C	45	45	45	45										
Lub oil temp. TC out	°C	64	63	64	65											
Engine media			MEAS	MEASURED												
	Lub oil temp ENGINE in	°C	45.0	Fuel oil press. ENGINE in		bar	7.80									
	Lub oil temp. THRUST SEGM.	°C	58.0	Fuel oil press. ENGINE out		bar	7.20									
	Lub oil press. SYSTEM OIL	bar	2.40	Fuel oil press. FILTER in		bar	7.6									
	Lub oil press. COOLING OIL	bar	2.40	Fuel oil press. FILTER out		bar	7.4									
	Lub oil press. TURB. OIL	bar	2.10	Hydraulic oil press. FILTER in		bar	2.3									
	Lub oil temp. TURB. OIL in	°C	45.00	Hydraulic oil press. FILTER out		bar	2.1									
	Lub oil temp. TURB. OIL out	°C	64.00	Hydraulic oil press. MAIN		bar	230.0									
	Axial vibration	mm	0.62	Sealing air press.		bar		PUMP 2	PUMP 3	PUMP 4	PUMP 5					
				Hydr. oil PUMP swash plate pos.		%	60.0	50.0	60.0							
Plane			MEAS													
	Exh. gas differential press. ECONOMIZER	mmWG	5.00													